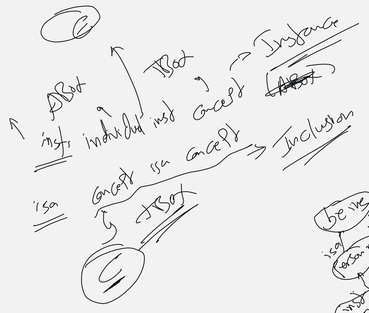
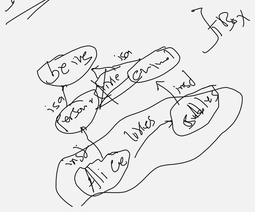


$\mathcal{E}_1 = \{likes\}$

$I(likes) = \{(Alice, Bubbles)\}$



Problem 10.1 (Modeling in Description Logic)

Consider the following situation:

- Some beings are persons, some are animals.
- Persons and animals may like other persons or animals.
- Alice is a person, and she likes the animal Bubbles.

1. Model this situation as a semantic network. Explain the different kinds of nodes and edges occurring in your network.
2. Model the same situation in first-order logic and compare the results.
3. Explain the difference between inst and is-a edges.
4. Explain the difference between having a relation edge between two concepts vs. asserting a relation between two individuals.

1) Nodes

- bees, Person, animal represent Concepts
- Alice, Bubbles represent individuals

2) Relations

Person isa bees, animal is a bees, {Person like animal, animal like Person}

Alice inst Person, Bubbles inst animal, Alice likes Bubbles

$\mathcal{E}_1 := \{bees, Person, animal\}$

$\mathcal{E}_0 := \{Alice, Bubbles\}$

$\mathcal{E}_2 := \{likes\}$

$\forall x. likes(x,y) \Rightarrow Person(x)$

$\forall y. likes(y,x) \Rightarrow animal(y)$

Axioms

$\forall x. Person(x) \Rightarrow bees(x)$

$\forall x. animal(x) \Rightarrow bees(x)$

$Person(Alice)$

$Animal(Bubbles)$

$likes(Alice, Bubbles)$

$\forall x. P(x)$

$V := \{ \quad \}$

$I(likes) = \{(Alice, Bubbles)\}$

Def 10.1. Semantic Network: A semantic network is a directed graph for representing knowledge:

- nodes represent **objects** and **concepts** (classes of objects).
- edges (called links) represent relations between these.

Def 10.2. Inclusion: We call links labeled by isa inclusion links or isa links (inclusion of concepts)

Def 10.3. Instance: We call links labeled by inst instance links or inst links (concept membership)

Def 10.4. Relations in Semantic Nets: We call all links which **except** isa and inst in a semantic network **relations**.

Def 10.5. Inference in Semantic Networks: Let N be a semantic network and R a relation in N such that $A \xrightarrow{R} B$, $A \xrightarrow{isa} C$ or $A \xrightarrow{inst} B$, $B \xrightarrow{isa} C$, then we can derive a relation $A \xrightarrow{R} C$ in N. The process of deriving new concepts and relations from existing ones is called **inference** and concepts/relations that are only available via inference are called **implicit** in a semantic network.

Example: Mar are derived relations, and je not allowed:

Figure 24: Inference in semantic networks

Def 10.6. Tree: We call the subgraph of a semantic network N spanned by the isa links and relations between **concepts** (in semantic nets) an **Tree**, or **Isa Hierarchy**.

Def 10.7. Graph: We call the subgraph of a semantic network N spanned by the inst links and relations between **objects** (in semantic nets) a **Graph**.

Example:

Figure 25: Tree and Graph in semantic networks

Def 10.25. $\mathcal{ALC}$ Syntax: The formulae of $\mathcal{ALC}$ are given by the following grammar:

$$F_{ALC} := C \mid \top \mid \bot \mid F_{ALC} \sqcap F_{ALC} \mid F_{ALC} \sqcup F_{ALC} \mid \exists R.F_{ALC} \mid \forall R.F_{ALC}$$

Syntax Examples:

- person $\sqcap$ $\exists$has_child.student$\}$

  This means the set of persons that have a child which is a student, i.e. parents of students

- person $\sqcap$ $\{\exists$has_child.$\exists$has_child.student$\}$

  grandparents of students.

Concept definitions Examples:

- man = person $\sqcap$ $\exists$has_chrom.Y_chrom
  meaning, a man is a person who has at lease one Y chromosome. person here is a primitive concept
- woman = person $\sqcap$ $\exists$has_chrom.Y_chrom
- mother = woman $\sqcap$ $\exists$has_child.person
- father = man $\sqcap$ $\exists$has_child.person
- grandparent = person $\sqcap$ $\exists$has_child.(mother $\sqcup$ father)

The complete $\mathcal{ALC}$ semantics:

- $[\![\top]\!] = \mathcal{D}$, $[\![\bot]\!] = \emptyset$, $[\![\mathcal{D}]\!] \subseteq \mathcal{D} \times \mathcal{D}$
- $[\![\neg\varphi]\!] = \mathcal{D} \setminus [\![\varphi]\!]$
- $[\![\varphi \sqcap \psi]\!] = [\![\varphi]\!] \cap [\![\psi]\!]$
- $[\![\varphi \sqcup \psi]\!] = [\![\varphi]\!] \cup [\![\psi]\!]$
- $[\![\exists R.\varphi]\!] = \{x \in \mathcal{D} \mid \exists y.\langle x,y\rangle \in [\![R]\!] \wedge y \in [\![\varphi]\!]\}$
- $[\![\forall R.\varphi]\!] = \{x \in \mathcal{D} \mid \forall y.\langle x,y\rangle \in [\![R]\!] \Rightarrow y \in [\![\varphi]\!]\}$
- $[\![\varphi \sqsubseteq \psi]\!] = T \Leftrightarrow [\![\varphi]\!] \subseteq [\![\psi]\!]$
- $[\![a \text{ R } b]\!] = T \Leftrightarrow \langle [\![a]\!], [\![b]\!]\rangle \in [\![R]\!]$

The complete translation:

$FOL_x(p) := p(X)$
$FOL_x(\bar{A}) := \neg FOL_x(A)$
$FOL_x(A \sqcap B) := FOL_x(A) \wedge FOL_x(B)$
$FOL_x(A \sqcup B) := FOL_x(A) \vee FOL_x(B)$
$FOL_x(A \sqsubseteq B) := FOL_x(A) \Rightarrow FOL_x(B)$
$FOL_x(A \equiv B) := FOL_x(A) \Leftrightarrow FOL_x(B)$
$FOL_x(\forall R.\varphi) := (\forall Y.R(X,Y) \Rightarrow FOL_y(\varphi))$
$FOL_x(\exists R.\varphi) := (\exists Y.R(X,Y) \wedge FOL_y(\varphi))$

For Formulae:

$FOL(A) := \forall X.FOL_x(A)$
$FOL(\sigma : \varphi) := FOL_w(\varphi)$
$FOL(a \text{ R } b) := R(a,b)$

Problem 10.2 (ALC)

Consider the following description logic signature

- *concept* symbols: i (for instructor), s (for student), c (for *course*), p (for program)
- *role* symbol m (for is-member-of) used for
  - *instructors* giving a *course*,
  - *students* taking a *course*
  - *students* being enrolled in a *degree program*
  - *courses* being part of a *degree program*

i	g	c
s	t	c
s	e	p
c	p	p

We use an extension of $\mathcal{ALC}$, in which there are dual roles: there is a role m^{-1} that captures the relation has-as-member, e.g., $MK\ m\ AI \equiv AI\ m^{-1}\ MK$.

1. For the *signature* above, give a *concept axiom* that captures that instructors can only be members of courses.
2. Give a *concept axiom* for the above *signature* that captures: *courses* that are taken by a *student* must be given by an *instructor*.
3. Calculate the translation to *first-order logic* of $s \sqsubseteq \forall m.\exists m.p$.
4. Given a *first-order model* $(\mathcal{D}, \mathcal{I})$, define an appropriate case of the *interpretation* mapping for the formula $\forall r^{-1}.C$.

$\{\forall x. \bar{c}\} = \{x \in D \mid \forall y. \langle x,y\rangle \in [\![r]\!] \Rightarrow y \in [\![\bar{c}]\!]\}$

$\forall x. \bar{c}\} = \{x \in D \mid \forall y. \langle x,y\rangle \in [\![r]\!] \Rightarrow y \in [\![c]\!]\}$

$\forall x. c\} = \{x \in D \mid \forall y. \langle x,y\rangle \in [\![r]\!] \Rightarrow y \in [\![c]\!]\}$

has_child

$\{(\forall x, \bar{c})\}$

$\{(\forall y, \bar{s})\}$

has_child

$\{(\forall x, \bar{c})\}$

has_member

$\{(\forall x, \bar{c})\}$

$x \rangle \in m^{-1}$

$y \rangle \in m$

$\forall$has_child.student

$\forall$has_child.$\bar{c}$

$\{\forall x. \bar{c}\} =$

$\{x \in D \mid \forall y. \langle x,y\rangle \Rightarrow c(y)\}$

$\Rightarrow$

$\{x \in D \mid \forall y. \langle x,y\rangle \in [\![r]\!] \Rightarrow y \in [\![\bar{c}]\!]\}$

$\{x \in D \mid \forall y. \langle x,y\rangle \in [\![r]\!] \Rightarrow y \in [\![c]\!]\}$

$\forall \cdot c = \{x \in D \mid \forall y. \langle x,y\rangle \in [\![r]\!] \Rightarrow y \in [\![c]\!]\}$

has_child

(x, y)

$y \rangle \in r^{-1}$

The complete translation:

$FOL_x(p) := p(X)$
$FOL_x(\bar{A}) := \neg FOL_x(A)$
$FOL_x(A \sqcap B) := FOL_x(A) \wedge FOL_x(B)$
$FOL_x(A \sqcup B) := FOL_x(A) \vee FOL_x(B)$
$FOL_x(A \sqsubseteq B) := FOL_x(A) \Rightarrow FOL_x(B)$
$FOL_x(A \equiv B) := FOL_x(A) \Leftrightarrow FOL_x(B)$
$FOL_x(\forall R.\varphi) := (\forall Y.R(X,Y) \Rightarrow FOL_y(\varphi))$
$FOL_x(\exists R.\varphi) := (\exists Y.R(X,Y) \wedge FOL_y(\varphi))$

For Formulae:

$FOL(A) := \forall X.FOL_x(A)$
$FOL(\sigma : \varphi) := FOL_w(\varphi)$
$FOL(a \text{ R } b) := R(a,b)$

$FOL(s \sqsubseteq \forall m.\exists m.p)$

$\forall x. FOL_x(s \sqsubseteq \forall m.\exists m.p)$

$\forall x. (FOL_x(s) \Rightarrow FOL_x(\forall m.\exists m.p))$

$\forall x. (s(x) \Rightarrow \forall Y. m(x,Y) \Rightarrow FOL_y(\exists m.p))$

$\forall x. (s(x) \Rightarrow \forall Y. m(x,Y) \Rightarrow \exists Z. m(Y,Z) \wedge FOL_z(p))$

$\forall x. (s(x) \Rightarrow \forall Y. m(x,Y) \Rightarrow \exists Z. m(Y,Z) \wedge P(Z))$

Problem 10.4 (ALC TBox)

Consider $\mathcal{ALC}$ with the following

- primitive concepts: woman, man
- roles: has_child, has_parent, has_sibling, has_spouse

Give an $\mathcal{ALC}$ *TBox* that defines the *concepts* person, parent, mother, father, grandmother, aunt, uncle, sister, brother, onlychild, cousin, nephew, niece, fatherinlaw, motherinlaw.

Person = man $\sqcup$ Woman

Person = Person $\sqcap$ $\exists$has_child.Person

mother = Woman $\sqcap$ Parent

grandmother = Woman $\sqcap$ $\exists$has_child.$\forall$parent

grandf = Woman $\sqcap$ $\exists$has_child.$\forall$parent

uncle = ——

sister = Woman $\sqcap$ $\exists$has_sibling.Person

$\forall$V.has

Person $\sqcap$ $\exists$has_child.$\forall$Person

Person $\sqcap$ $\sqcap$ Br $\cup \Rightarrow \cup$ Br

fatherinlaw

Problem 10.3 (ALC Semantics)

Consider the $\mathcal{ALC}$ *concepts* $\forall R.(C \sqcap D)$ and $\forall R.C \sqcap \forall R.D$.

1. By applying the semantics of $\mathcal{ALC}$, show that the two are equivalent.
2. Translate both formulas to first-order logic and state which FOL formula we would need to prove (e.g., with the ND calculus) to show that the two are equivalent.

$\{\forall R.(C \sqcap D)\} = \{x \in D \mid \forall Y. \langle x,y\rangle \in [\![R]\!] \Rightarrow y \in [\![C \sqcap D]\!]\}$

$\Rightarrow y \in ([\![C]\!] \cap [\![D]\!])$

$\{\forall R.C \sqcap \forall R.D\} = \{\forall R.C\} \cap \{\forall R.D\}$

$= \{x \in D \mid \forall Y. \langle x,y\rangle \in [\![R]\!] \Rightarrow y \in [\![C]\!]\} \cap$

$= \{x \in D \mid \forall Y. \langle x,y\rangle \in [\![R]\!] \Rightarrow y \in [\![D]\!]\}$

$\{x \in D \mid \forall Y. \langle x,y\rangle \in [\![R]\!] \Rightarrow y \in ([\![C]\!] \cap [\![D]\!])\}$